

The Notebook You Read Is Not the Notebook That Ran: Execution-Order Reconstruction for LLM Understanding of Jupyter Notebooks

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Abstract

Context: A Jupyter notebook stores cells in *visual* order, which often differs from the *execution* order they actually ran in. Its JSON format also buries code under base64 images, duplicate MIME formats, and metadata. Stripping content saves tokens but also loses clues needed to recover true execution order. **Objective:** We test whether reconstructing execution order improves an LLM’s ability to track program state, both in controlled and real settings. **Method:** We built JupPreprocessor, a serializer that never calls an LLM: it resolves what it can recover statically and flags what it cannot. We evaluated it on a synthetic benchmark spanning fifteen models and seven vendors, and on ground truth from re-executing 112 real crashed notebooks (JunoBench) under both orderings. **Results:** Unaided, models resolve state correctly 0%–97% of the time; our fix scores 100% at roughly 95% fewer tokens. On 52 real order-divergent variables, however, our serializer shows no per-item accuracy advantage over the raw file for gpt-4o, our one answer model on this corpus, once it answers (23–27% correct, overlapping 95% confidence intervals), a result that held after we fixed a grading bug. The defensible advantage is narrower: the raw file is too large to fit in context for 62% of these items, versus 15–17% for the compressed formats. **Conclusions:** Execution-order reconstruction is necessary and sufficient for state-tracking in a controlled setting. On real notebooks, our evidence shows only that the serializer keeps more notebooks readable, not that answers become more accurate once a file is readable.

Keywords: Jupyter notebooks; large language models; execution-order reconstruction; program state serialization; empirical software engineering; notebook reproducibility

1 Introduction

Ask a large language model to debug a Jupyter notebook, and it will confidently reason about a version of events that may never have happened. This is because the `.ipynb` file format records cells in the order they visually appear on screen, not the order they were actually executed in. A user can run cell three, then cell one, then delete cell two entirely, and the saved file will still list the cells top to bottom as if nothing unusual occurred. The only surviving trace of what really happened is a small integer attached to each cell, called `execution_count`, which simply records the number of times the notebook’s kernel has executed something by the time that cell last ran. As we show later in this paper, models frequently fail to use this signal correctly even when it is sitting right there in the file (Sect. 5.2).

This is not a rare edge case. A prior large-scale study of GitHub notebooks found that roughly 36% of them execute their cells in a non-linear order (Siddik et al., 2025). We wanted to know how bad the problem is specifically for the population an LLM assistant is most likely to encounter: notebooks that were saved right after a crash, where a user was actively debugging something and did not have time to clean up. We found the picture is considerably worse than the general GitHub average: **59.8% of these notebooks are saved out of order, 89.3% show gaps consistent with cells being re-run or deleted, and 92.0% carry at least one of these warning signs** (Sect. 5.3).

We organize our empirical investigation around five research questions, each answered in a specific part of the paper rather than left implicit:

- **RQ1.** How prevalent are non-linear execution and other state hazards in real, crashed machine-learning notebooks, the population most likely to be handed to an LLM for debugging? (Sect. 5.3)
- **RQ2.** Does execution-order reconstruction measurably improve LLM state-tracking accuracy on a fair, context-bounded synthetic benchmark, isolated from mere retrieval-under-noise and from token-count confounds? (Sect. 5.2)
- **RQ3.** Does this synthetic-benchmark advantage transfer to real, messy notebooks, when measured against executable ground truth rather than assumed correctness? (Sect. 5.4–5.5)
- **RQ4.** What is the token-cost and robustness profile of the preprocessing pipeline on real notebooks, and does adding state-safety annotations erode the token savings? (Sect. 5.1)
- **RQ5.** Where does the approach fail or reach its limits, on already-clean data, in the preprocessor’s own core execution-order-sort mechanism, and relative to alternative architectures such as reactive notebooks? (Sect. 5.6, Sect. 6, Sect. 7)

Answering these five questions is also how this paper makes its four main contributions:

1. **A three-tier treatment framework.** We separate what a preprocessing tool can actually *resolve* on its own from what it can only *detect and flag* because the true value was simply never written down anywhere in the file, and from what genuinely requires *running the code* to know for certain.
2. **JupPreprocessor**, an implementation of the first two tiers that never calls an LLM and never executes any code itself. It adds six lightweight annotations to a notebook (a history of how each variable was mutated over time, the last value a deleted “ghost” variable was seen holding, a flag for code whose output may not be reproducible, whether an error was later worked around or was fatal, inferred type and shape information, and a flag for a specific weakness in its own ordering logic that we discovered and describe below, affecting 26.6% of real notebooks). All six together cost well under 1% in extra tokens.
3. **A fair benchmark we call Notebook-NIAH.** It is built so that the baseline condition always fits inside the model’s context window; running out of room is recorded as its own outcome, never scored as a wrong answer. It includes a compression-only control, a comparison between a “static” task (plain retrieval) and a “stateful” task (state reconstruction) to separate the two skills, component-by-component ablations, and a capability curve spanning fifteen models across seven vendors.

4. **Executable ground truth on real notebooks.** By actually re-running out-of-order notebooks under both their visual and their true execution order, we surface real variables whose final value differs between the two readings, in effect showing that the model a user thinks they trained is not always the model they actually trained. This gives us the first real-notebook state-accuracy measurement of its kind, together with an honest account of what still does not work.

To our knowledge, this is the first paper to treat notebook serialization format itself as a controlled experimental variable, holding the answer model and prompt fixed and varying only how the notebook is encoded, and to evaluate that variable specifically on crashed, out-of-order notebooks rather than the cleaner, curated corpora used elsewhere in this literature (Sect. 7 contrasts this directly against the closest prior systems).

2 Root Causes and the Coupling Problem

We group the problems that make a `.ipynb` file hard for an LLM to understand into ten root causes, six about program state and four about token cost. The state causes are: (A) non-linear execution, meaning the saved order does not match the run order; (B) in-place mutation, where an operation changes an object without leaving any static trace of the change; (C) ghost variables, names that are still referenced later in the notebook even though the cell that created them was deleted; (D) re-execution divergence, where re-running a cell with a different random seed, a changed input file, or a live network call would produce a different result than what is stored; (E) silenced errors, where a traceback is visible but a later cell shows the user worked around it successfully; and (F) type erasure, where an output is stored as plain text and the reader can no longer tell whether it represents a number, a table, or an image description. The token causes are: (G) the same output being duplicated across several MIME formats (for example, a table stored simultaneously as plain text, HTML, and LaTeX); (H) images encoded as long base64 text strings that a model cannot see but must still pay to read as tokens; (I) old outputs from earlier runs of a cell being kept alongside the current one; and (J) the sheer verbosity of nested JSON structure itself. These two groups pull against each other: restoring missing state information costs tokens, and cutting tokens tends to remove exactly the state information a model would need.

Three of the six state causes cannot be solved by any amount of clever static analysis, because the information they concern was never written into the file in the first place. A cell that mutates an object in place (B) leaves no record of what the object looked like afterward. A deleted cell that created a ghost variable (C) leaves no record of what created it. A cell that behaves differently on re-execution (D) leaves no record of what the alternative outcome would have been. Table 1 turns this observation into a concrete design rule: our tool should not pretend to have solved something it cannot actually know. The key idea behind our second tier is that a model does not need the missing value in order to avoid a wrong answer; it only needs to know that the value is missing, and roughly where the uncertainty entered the notebook. Once a model is told “this could be wrong,” it reasons more carefully than it would given no warning at all.

3 JupPreprocessor

JupPreprocessor is a single-pass, deterministic transform written in Python. It never calls an LLM, never makes a network request, and never attaches to a Python kernel; it only reads the static text of the notebook file. Given a raw `.ipynb` file, it produces a YAML document containing

Tier	Treatment	Causes	Mechanism
1	Resolve	A, G–J	Reorder by <code>execution_count</code> ; strip MIME/base64/metadata; latest output only; YAML
2	Detect & flag	B–F	Mutation timeline; ghost last-observed value; divergence-risk flags; tolerated/terminal errors; typed shape hints
3	Runtime verify	B–E fully	Sandboxed re-execution diffed against static predictions

Table 1: The three-tier treatment framework.

an `execution_order` field that lists the cells in the order they were actually run, not the order they appear on screen, with markdown cells re-attached to whichever code cell they originally preceded. Each cell is annotated with an `error_status` (was any error tolerated or fatal) and a `divergence_risk` flag (does this cell contain code, such as an unseeded random draw or a network call, whose output might change on re-execution). Alongside the cells, the tool produces a `variables` table recording, for every name it can track: its inferred type, its shape where one can be inferred, the cell that first defined it, the cell that most recently touched it, a short history of how it was mutated, a status of either `active` or `ghost`, and a confidence level for all of the above. If a cell contains something the Python parser cannot handle on its own, such as an IPython magic command or a shell escape, the tool falls back to a simpler regular expression parser for that cell rather than giving up on the notebook entirely. We chose YAML over the notebook’s native JSON purely for compactness: the same structural information costs noticeably fewer tokens once repeated key names and bracket nesting are removed.

Sorting by `execution_order` is not always as clean as it sounds, and it is worth flagging the core limitation here rather than only in Sect. 6. Because `execution_count` records only a cell’s *most recent* run, a cell re-run late in a session, for example an `import` statement re-executed after a kernel restart, can end up sorted *after* code that already depends on it, producing an order that could never actually be replayed in a fresh kernel. We found a real example of exactly this in one of our own ground-truth notebooks; the sixth annotation JupPreprocessor emits, `execution_order_risk`, exists specifically to flag cells affected by this pattern rather than silently sorting them into a plausible-looking but wrong position. Sect. 6 gives the full worked example and quantifies how often this affects real notebooks.

3.1 Use of large language models and AI agents in this study

Following this journal’s authorship policy, we document our use of LLMs and AI agents explicitly instead of treating it as an incidental detail. LLMs play two different roles in this work, and we keep them clearly separated throughout the paper. First, they appear as *answer models under test*: the fifteen models evaluated on the synthetic benchmark, and gpt-4o on the real-notebook evaluation, are the subject of our measurements, not a tool we used to produce those measurements. Second, they appear as *research and drafting assistance*. An automated grading model (gpt-4o-mini) was used to classify free-text answers against a small set of stored ground-truth candidates in the real-notebook evaluation (Sect. 5.4); we independently audited its reliability and report the bug we found in it in full (Sect. 5.5) instead of treating it as an unremarkable piece of tooling. An LLM-based coding and research agent was also used throughout the project to write the experiment scripts, run the measurements reported here, draft prose from the resulting reports, and carry out

an independent verification pass that cross-checked every quantitative claim in this manuscript against the source report or script output that produced it (the verification log is kept in the project repository; see the Data Availability statement). No LLM is credited as an author, in line with this journal’s position that an LLM cannot satisfy authorship criteria. The corresponding author takes full responsibility for the accuracy of every claim made here, having directed and reviewed every experiment and every correction described in this manuscript. That includes the several occasions, described below, on which an initial LLM-assisted result turned out to be wrong: the judge bug, the `execution_count`-sort edge case, and the mistaken original explanation of the Themisto result. Each of those was corrected in this manuscript instead of being published as first found.

4 Experimental Design Principles

We designed every experiment in this paper around five principles, listed here because they explain why several early, more favorable-looking results were later revised downward. First, **fairness**: whenever a prompt is too long to fit in a model’s context window, we record this as a separate outcome we call `OVERFLOW`, never as an incorrect answer; a model is only marked wrong if its prompt actually fit and it still answered incorrectly. Second, **controls**: we include a condition called `plain_strip`, which applies the same amount of compression as our full pipeline but keeps the cells in their original visual order and adds no state table, so that we can separate the effect of simply shrinking the prompt from the effect of restructuring it; two further ablated conditions, `reorder_only` and `table_only`, isolate each individual mechanism. Third, a **static-versus-stateful contrast**: the identical needle string is used in both settings, and only the stateful setting reassigns that value out of order behind a visually later decoy, so that any gap seen only in the stateful setting can be attributed to state reconstruction and not to plain retrieval. Fourth, **determinism**: the synthetic haystacks are generated from CRC32-seeded pseudorandom sequences, and the dual-order re-execution of real notebooks uses seeded random-number generators throughout. Fifth, **honest reporting**: we state negative results and by-construction caveats directly alongside the results they qualify, instead of deferring them to a separate section a reader might skip.

5 Results

All real-notebook results in this section draw from a single, specific population: real, previously crashed machine-learning notebooks from JunoBench. This population was chosen deliberately, not for convenience: notebooks saved after a crash are exactly the ones a debugging assistant is most likely to be handed, and Sect. 5.3 shows they are also far more likely to be out of order than the general population of GitHub notebooks. Of the 112 original JunoBench notebooks, 67 are non-monotonic (saved out of visual order); of those, 62 could be executed usefully in our dual-order sandbox, yielding the 52 order-divergent variables across 16 notebooks used throughout Sect. 5.4–5.5. We return to this funnel, and to what it does and does not license us to claim about notebooks outside this specific population, in Sect. 6.

5.1 Compression and robustness on real notebooks

We ran JupPreprocessor over 40 real crashed machine-learning notebooks drawn from JunoBench (Wang et al., 2025) (seed 42). All 40 were preprocessed without error, and the aggregate token count dropped by 96.4%, with a median per-notebook reduction of 91.2%. The single largest case shrank from about 7.5 million tokens to roughly 11,200. One already-lean notebook actually *grew*

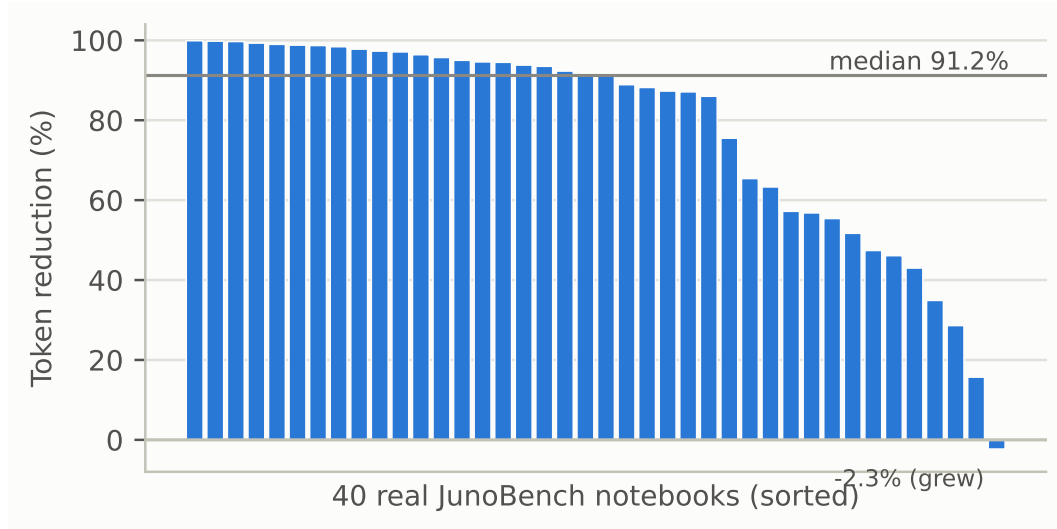


Figure 1: Preprocessing shrinks real notebooks by a median of 91.2%, with a single, well-understood exception. Per-notebook token reduction on 40 real JunoBench notebooks, sorted. The single negative case is an already-lean notebook that grows under YAML plus the state table.

by 2.3% once our YAML structure and state table were added; we call this the “already-clean” failure mode, and it occurred in exactly one of the forty notebooks we tested. Adding all six state-safety annotations described earlier, including the `execution_order_risk` flag introduced in Sect. 6, together costs well under 1% in extra tokens (Figure 1): the median reduction moved only from 91.5% to 91.2% once that sixth annotation was included.

5.2 Fair Notebook-NIAH: retrieval is not the failure mode; state is

Notebook-NIAH is our adaptation of the “needle in a haystack” testing idea used elsewhere to probe long-context recall: we hide a single target value somewhere inside a long, synthetically generated notebook, and ask the model to report it back. We swept three depths (0.1, 0.5, and 0.9 of the way through the file) crossed with four lengths (roughly 8,000, 30,000, 60,000, and 100,000 tokens), all kept under a 126,000-token cap, with three trials per grid cell (one trial per cell for the two 2026-03 model snapshots). In the **static** setting, the needle is placed once and never reassigned; here gpt-4o scored 100% for every condition at every combination of depth and length, which tells us that noise alone does not impair retrieval once a prompt actually fits in context, correcting a claim we had assumed was true when we first started this project. The **stateful** setting is the one that actually matters for this paper’s argument: here, the needle’s true final value has the highest `execution_count` but appears *earlier* in the notebook’s visual layout, while a decoy value sits later in the visual order with a lower `execution_count`, exactly mimicking the real-world hazard described in Sect. 2. We tested fifteen models across seven vendors on this stateful setting: the nine OpenAI models ran the full 36-trial grid, and six non-OpenAI models (NVIDIA Nemotron-3-Super-120B, MiniMax-M3, Cohere North-Mini-Code, InclusionAI Ling-3.0-Flash-Fin, Poolside Laguna-S-2.1, and Zhipu GLM-5.3-Flash) ran a reduced grid of one length by three depths by one or two trials by three conditions, sized to fit within each provider’s free-tier rate limit. The results appear in Figure 2: unaided (**raw**) accuracy ranges from 0% for MiniMax-M3 and Cohere North-Mini-Code up to 97% for gpt-5 on the full grid. GLM-5.3-Flash scored 100% on its much smaller sample of only three trials, which is too little evidence to treat as a reliable ceiling rather

than a lucky run. The compression-only control, `plain_strip`, stayed close to 0% throughout. Our full pipeline, `ours`, scored 100% on every single model. The component ablations on gpt-4o make the mechanism clear: reordering alone (`reorder_only`) also reaches 100%, while the state table alone with no reordering (`table_only`) scores 0%, the same as `plain_strip`.

We draw five conclusions from this curve. First, reassigning a value out of order degrades every model tested, even though the metadata needed to resolve it correctly is sitting right there in the JSON; the failures concentrate at the deepest positions in the file. Second, moving from gpt-4o-mini to gpt-5 (a span of three OpenAI generations) took unaided accuracy from 3% to 97%, but never closed the small remaining gap that a deterministic serializer closes completely on every model at once, using roughly 95% fewer tokens in the process. In other words, model scaling narrows the gap; it does not eliminate it the way reordering does. Third, compression by itself does not help, and adding metadata without also fixing the order does not help either: execution-order reordering is both necessary and sufficient for this task on its own. Fourth, accuracy does not increase smoothly with either model tier or release date: gpt-4.1-nano actually outperforms the ostensibly stronger gpt-4.1-mini, and the newer 2026-03 gpt-5.4-mini snapshot scores below the earlier gpt-5-mini, so relying on serialization is a more predictable lever than simply picking a newer model. Fifth, five of the six non-OpenAI models reproduce the same pattern seen across the nine OpenAI models: MiniMax-M3’s 0% raw score is the single worst result in the entire curve, Nemotron-3-Super’s 50% sits in the middle of the pack, and all six non-OpenAI models are fixed completely by `ours` once it is applied. Because each non-OpenAI model only received three to six trials on a reduced grid, we treat this as a spot check and not a demonstration of broad cross-vendor generality: it tells us that five of the six additional vendors did not contradict the main pattern, and that the sixth (GLM-5.3-Flash) is too thinly sampled to draw any conclusion from either way. We should also be transparent about a limitation of the benchmark itself: because it is built to instantiate exactly the defect JupPreprocessor is designed to fix, a perfect `ours` score mainly confirms that the transform behaves correctly for that specific defect, not that models generally prefer our output format in every setting. The evidence we treat as load-bearing is the shape of the raw capability curve together with the ablations, not the `ours` = 100% figure taken on its own.

Table 2 gives the exact per-model numbers underlying Figure 2, since a figure alone can be hard to read precisely for fifteen separate models.

5.3 State hazards are the norm in the debugging population

We scanned all 112 original JunoBench notebooks for the two static warning signs a preprocessor can actually detect without executing anything: a mismatch between visual and execution order, and gaps in the `execution_count` sequence that indicate a cell was re-run or deleted. We found that 59.8% of these notebooks are non-monotonic (their visual order does not match their execution order), 89.3% show execution-count gaps (a precondition for the ghost-variable problem described in Sect. 2), and 92.0% show at least one of these two hazards. Compared with the roughly 36% figure reported for GitHub notebooks in general (Siddik et al., 2025), this suggests that the specific notebooks most likely to be handed to an LLM for help are also the ones most likely to mislead it.

5.4 Executable ground truth on real notebooks

To find out whether real notebooks actually contain the kind of order-divergent variable our synthetic benchmark is built to test, we took every out-of-order notebook we could identify and executed it twice inside an isolated sandbox, once sorted by `execution_count` and once in its original visual order, using identical random seeds and per-cell timeouts for both runs. We first installed

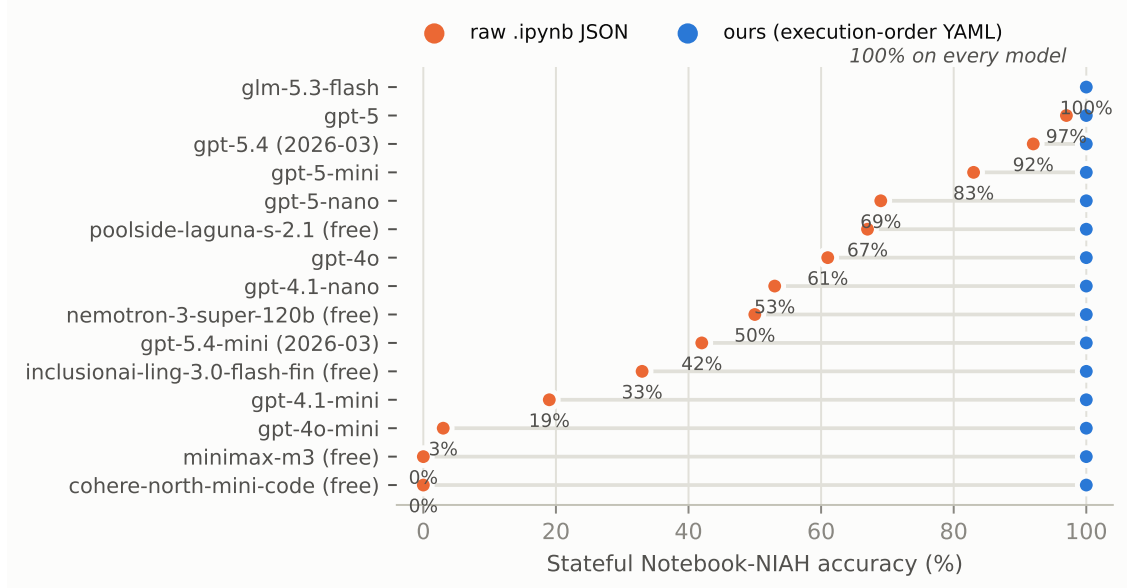


Figure 2: Execution-order reordering closes the state-tracking gap completely on every model, while unaided accuracy varies enormously and non-monotonically with model choice. Stateful Notebook-NIAH accuracy across fifteen models: nine OpenAI models spanning four generations, plus six non-OpenAI models (NVIDIA Nemotron-3-Super-120B, MiniMax-M3, Cohere North-Mini-Code, InclusionAI Ling-3.0-Flash-Fin, Poolside Laguna-S-2.1, Zhipu GLM-5.3-Flash) on a reduced grid, each via its own free tier. Unaided (**raw**) accuracy spans 0% to 97% on every adequately-sampled (full-grid) model and never reaches the 100% that execution-order reconstruction (**ours**) achieves on every model; one thinly-sampled exception (GLM-5.3-Flash, $n = 3$) also scored 100% raw, too small a sample to draw a conclusion from. **plain_strip** (not shown) is $\approx 0\%$ throughout.

`torch/torchvision` (CPU wheels, Python 3.14), and later widened the corpus further using a separate Python 3.12 virtual environment with TensorFlow, since no TensorFlow wheel exists for Python 3.14 at the time of writing. This let 62 of 67 candidate notebooks execute usefully, yielding 52 order-divergent variables across 16 notebooks. This is a conservative lower bound: several notebooks still could not fully execute because of missing data files or heavy dependencies, one TensorFlow notebook remains unexecutable because a large mid-execution network download races against our per-cell timeout (a genuine flaky-network issue rather than a missing-dependency one), and we observed run-to-run variance of 52 to 55 items across repeated extraction runs, traced to other network-dependent code inside the notebooks themselves. We found two distinct kinds of divergence. The first is *value divergence*: in `sklearn_9`, the model that actually exists at the end of the session is a plain `LogisticRegression()`, while a naive visual-order reading would conclude it is `LogisticRegression(solver='liblinear')`. The second is *existence divergence*, which introduces what we call **phantom variables**: the mirror image of a ghost variable, in that the code defining it is visible in the file but the definition itself never actually completed successfully during the real session. Table 3 breaks the 52 items down by notebook and divergence type: the distribution is uneven by design rather than by selection, since some notebooks simply contain far more order-divergent variables than others once actually executed.

We then asked gpt-4o to report the final state of each of the 52 items under three serializations, and graded its answers with a three-way judge: TRUTH means the answer matches the execution-order state, DECOY means it matches the (wrong) visual-order state, and NEITHER means it matches

Model	raw	plain_strip	ours
gpt-4o	61% (22/36)	0% (0/36)	100% (36/36)
cohere-north-mini-code (free)	0% (0/3)	0% (0/3)	100% (3/3)
glm-5.3-flash	100% (3/3)	33% (1/3)	100% (3/3)
gpt-4.1-mini	19% (7/36)	0% (0/36)	100% (36/36)
gpt-4.1-nano	53% (19/36)	8% (3/36)	100% (36/36)
gpt-4o-mini	3% (1/36)	0% (0/36)	100% (36/36)
gpt-5	97% (35/36)	8% (3/36)	100% (36/36)
gpt-5-mini	83% (30/36)	0% (0/36)	100% (36/36)
gpt-5-nano	69% (25/36)	0% (0/36)	100% (36/36)
gpt-5.4 (2026-03-05)	92% (11/12)	0% (0/12)	100% (12/12)
gpt-5.4-mini (2026-03-17)	42% (5/12)	0% (0/12)	100% (12/12)
inclusionai-ling-3.0-flash-fin (free)	33% (1/3)	0% (0/3)	100% (3/3)
minimax-m3 (free)	0% (0/6)	0% (0/6)	100% (6/6)
nemotron-3-super-120b (free)	50% (3/6)	0% (0/6)	100% (6/6)
poolside-laguna-s-2.1 (free)	67% (2/3)	0% (0/3)	100% (3/3)

Table 2: Exact per-model accuracy on the stateful Notebook-NIAH benchmark. The nine OpenAI models ran the full 36-trial grid; the six non-OpenAI models ran a reduced grid (3–6 trials) sized to each provider’s free-tier rate limit. **ours** is 100% on all fifteen models with no exception. Compiled automatically from per-model reports; see `reports/model_curve_report.md`.

neither stored candidate. The results, after fixing a judge bug we describe below, appear in Figure 3, together with 95% Wilson confidence intervals (a standard way of expressing our uncertainty about a proportion measured from a limited number of trials) on the overall-truth rate: **raw** 23% [14, 36], **plain_strip** 27% [17, 40], **ours** 27% [17, 40]. We report four findings here, arrived at only after three separate rounds of self-correction. The first and most robust finding, unchanged by any of those corrections and if anything strengthened by them, concerns simple **access**: the raw JSON file is too large to fit in context for 62% of these real divergent items at $n = 52$, up from 51% at an earlier $n = 41$ snapshot, since larger real notebooks kept entering the corpus as we widened it; our serializer reaches 85% of items and **plain_strip** reaches 83%. The second finding is less flattering: once an answer is actually given, we find *no* distinguishable accuracy advantage for **ours**. It is tied with **plain_strip** at 27% overall truth, raw trails only nominally at 23%, and all three confidence intervals overlap heavily. This matches the conclusion we already reached at $n = 41$ after fixing the judge bug, so the larger sample confirmed the finding instead of reversing it.

The third finding is about the correction process itself, and we consider it worth reporting as a finding in its own right, broken here into its four separate stages.

Initial pilot. An initial 13-item pilot looked like a clean win for our method, at 6 out of 13 correct versus raw’s 3 out of 13. Expanding to 41 items narrowed this gap to something indistinguishable from noise.

Judge bug and bias. A manual audit of the automated gpt-4o-mini judge, in which we independently re-checked 25 of 123 answer rows by hand (Sect. 5.5), then uncovered a 24% disagreement rate concentrated in a single, specific mechanism: whenever one of the two stored ground-truth candidates was “the variable was never defined,” the judge tended to credit any other concrete-looking answer to the remaining candidate without actually checking whether the value matched. This bug happened to favor our method’s tendency to produce a confident, specific-looking guess from its variable table, even when that guess was wrong.

Post-fix result. Once we rewrote the judge’s prompt to require an explicit literal or close value

Notebook	Items	Value divergence	Existence divergence
NBspecific.10.ipynb	1	0	1
NBspecific.11.ipynb	2	0	2
NBspecific.15.ipynb	1	1	0
matplotlib.1.ipynb	4	4	0
numpy.1.ipynb	12	0	12
numpy.11.ipynb	8	1	7
pandas.3.ipynb	5	0	5
sklearn.7.ipynb	1	0	1
sklearn.9.ipynb	3	1	2
torch.10.ipynb	1	1	0
torch.11.ipynb	1	1	0
torch.3.ipynb	6	1	5
torch.4.ipynb	1	0	1
torch.6.ipynb	3	0	3
torch.7.ipynb	1	1	0
torch.8.ipynb	2	0	2
Total (16 notebooks)	52	11	41

Table 3: The 52 real order-divergent variables by notebook and divergence type. Existence divergence (phantom variables) accounts for the large majority (41 of 52); value divergence, where a variable exists under both readings but holds a different value, accounts for the remaining 11. `numpy.1.ipynb` alone contributes 12 items, all existence divergence, and is the notebook responsible for the `execution_count`-sort edge case discussed in Sect. 6.

match, our method’s apparent advantage disappeared entirely.

Expanded corpus confirmation. Expanding the corpus a second time, to 52 items via the newly built TensorFlow environment, reproduced this corrected, more modest result instead of reversing it again. We report all four stages of this progression, not only the final number, because each successive correction moved the conclusion in the same direction: toward a more modest and, we believe, more trustworthy claim.

5.5 Judge validation

To check the automated judge’s reliability directly, a member of our project team manually re-graded 25 of 123 answer rows by comparing each sampled answer against the stored ground-truth candidates by hand, without invoking any LLM classifier in that re-check. The 25 rows were a stratified random sample (seed 42), not a sample selected after looking for suspected problem cases: we fixed the random seed before drawing the sample, so the audit could not have been enriched with rows we already suspected were wrong. This produced 19 out of 25 (76%) agreement with the original automated judge. All six disagreements shared one underlying cause, which we can demonstrate with a single reproducible example: for `numpy.1.ipynb`’s `device` variable, where the truth candidate was “never defined” and the decoy candidate was the literal string `cpu`, the identical answer text `cuda` was graded DECOY under one serialization condition and TRUTH under another. Neither grade was actually correct, since `cuda` matches neither stored candidate. We rewrote the judge’s prompt to require an explicit literal or close value match, and to default to NEITHER whenever this is uncertain. We also began persisting the model’s full, untruncated answers for future audits (`tests/real_state_eval_raw_answers.json`). Full detail is available in `reports/judge_validation_report.md`.

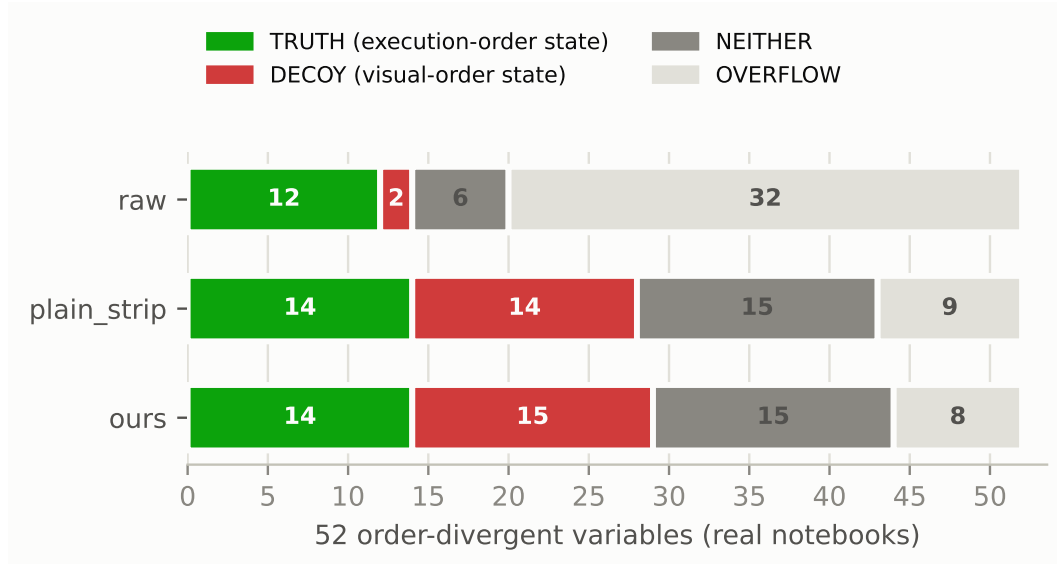


Figure 3: Raw JSON’s biggest problem on real notebooks is fitting in context at all, not accuracy once it does: OVERFLOW dominates its outcome bar, while `ours` and `plain_strip` reach far more items without a matching gain in TRUTH. Outcomes on the 52 real order-divergent variables per serialization. OVERFLOW = the prompt could not fit the 126k context cap.

Post-fix confirmation. A prompt fix is only as trustworthy as the next independent check of it, so we repeated this audit on a second, non-overlapping sample after applying the fix: 25 further rows, stratified by condition, drawn with a different random seed (7) from the corrected-judge dataset. Agreement rose from 76% to 92% (23 of 25). The two remaining disagreements were both the same underlying variable (`pandas.3.ipynb`’s `resultados`), where the judge credited a fully-populated DataFrame answer as matching a stored decoy candidate that is literally an *empty* DataFrame with matching column names but no rows, a narrower, DataFrame-specific failure mode distinct from the null-candidate bug described above. We report this as a residual limitation rather than a clean bill of health: readers should continue to treat the exact TRUTH/DECOY/NEITHER counts in Sect. 5.4 as approximate, with the qualitative conclusion, no detectable accuracy advantage at this scale, being what we actually stand behind rather than any single precise tally. Full detail is in `reports/judge_post_fix_validation_report.md`.

5.6 Honest negatives

On the Themisto benchmark (Grotov and Titov, 2025), whose notebooks were already cleaned of HTML, base64 content, and JSON metadata by its own authors, applying our preprocessing actually *reduced* output-prediction accuracy, from 40.0% down to 26.7%, while also adding token overhead rather than removing it. We did not simply accept the published explanation for this at face value; instead, we measured the actual breakdown of where the errors occurred, and found the original explanation was wrong. Of the items our method answered incorrectly, most (9 of 11) were *also* answered incorrectly under the raw condition, because both formats were guessing at data that neither one actually had access to; only 2 of the 15 items in this sample decided the entire gap between the two conditions. In both of those two decisive items, the correct value was present verbatim in our preprocessed prompt; it had not been stripped or lost anywhere. What we found instead, by measuring character distances directly, is that the variable-state table our pipeline

inserts between the notebook’s code history and the final query pushes the correct evidence 2.7 to 7.7 times farther from the point where the model must answer than it sits in the raw format, and in both cases the model’s wrong answer matched a closer but incorrect value shown elsewhere in the same prompt. This looks like a lost-in-the-middle effect, where information further from the point of decision is used less reliably, and not an information-loss effect where the value was simply missing. Structure is a net negative specifically in a setting like this one, where there was no token bloat to strip in the first place, so adding structure only adds distance without buying back any savings. Separately, on a full-context JunoBench debugging check, accuracy was identical (8 of 8 correct under both conditions) while our version used 79.1% fewer tokens: a cost result, not an accuracy result.

We want to state plainly what this Themisto result does and does not mean for the method as a whole. It does not mean the method generally hurts accuracy: on Themisto specifically, there was no token bloat to remove in the first place, so the only thing our added structure could do was push evidence farther from the query, which is exactly what happened. Our method’s actual strength lies in bloated, scrambled notebooks like the JunoBench corpus used throughout this paper, not in notebooks that are already lean and already in the correct order. A practitioner deciding whether to apply JupPreprocessor to a given notebook should weigh how bloated and how out-of-order it actually is, not assume the tool is a uniform improvement regardless of input.

6 Threats to Validity

6.1 Construct validity

The Notebook-NIAH synthetic benchmark is, by its own construction, favorable to our method: it deliberately instantiates the exact defect (a value reassigned out of order) that JupPreprocessor is designed to fix, so a perfect score on **ours** mainly confirms that the transform is correct and complete for that specific defect. It does not by itself show that language models generally prefer our output format in other settings. To be explicit about the boundary of this claim: Notebook-NIAH tests exactly one of the six state hazards named in Sect. 2, non-linear execution (cause A). It does not instantiate in-place mutation (B), ghost variables (C), re-execution divergence (D), silenced errors (E), or environment-dependent branching, and we do not claim that “execution-order reconstruction is necessary and sufficient” extends to any of those other hazards; it is a claim about this one benchmark’s one failure mode, not about notebook state tracking in general. We treat the benchmark’s controls (`plain_strip`, `reorder_only`, `table_only`) together with the independent real-notebook study as the evidence that actually matters here, not the **ours** = 100% figure taken in isolation.

The automated LLM judge used for real-notebook grading (gpt-4o-mini) had a measured 24% disagreement rate with independent manual re-grading before we found and fixed the bug behind it (Sect. 5.5); readers should treat the exact TRUTH and DECOY counts as approximate even after the fix, since no fully human-annotated ground truth exists for this task.

`execution_count`-sort itself also has a real, verified construct-validity problem of its own: this field only records a cell’s *most recent* run, not a mutually-consistent history of the whole session, so an early cell that gets re-run late (for example, an import statement re-executed after a kernel restart) can end up sorted *after* code that already depends on it. The resulting order is not something you could actually replay in a fresh kernel, and it does not correspond to any real moment in the notebook’s history. We verified this directly in `numpy_1.ipynb`, where an import cell carries `execution_count=27` while a cell that needs `torch` carries `execution_count=2` (see `reports/execution_count_reordering_limitation.md`); our new `execution_order_risk` anno-

tation correctly fires on exactly the 6 cells affected by this issue. An earlier, smaller snapshot of the ground-truth corpus ($n=41$, before we expanded it to $n=52$ using the Python 3.12 and TensorFlow environment) estimated that this mechanism plausibly explained 8 of those 41 items, 7 from `numpy_1.ipynb` and 1 from `torch_4.ipynb`, rather than genuine out-of-order-execution divergence. We have not re-quantified this specific estimate at $n=52$, so we flag it here as an estimate carried over from an earlier snapshot of the corpus rather than a current, re-verified count. Both our tool and the ground-truth methodology inherit this same underlying assumption regardless. A separate, corpus-wide static scan that cost nothing and made no LLM calls (`reports/execution_count_consistency_report.md`), restricted to names bound by an `import` statement for a cleaner signal than a noisier count dominated by ordinary reuse of short generic variable names, found that this issue affects **29 of 109 analyzable JunoBench notebooks (26.6%)**, which is common enough to be worth fixing directly. The preprocessor now does exactly that: a new `execution_order_risk` per-cell annotation, verified against the 6 affected cells in `numpy_1.ipynb`, passes the full existing unit test suite, and costs a negligible amount of extra tokens. We chose this detect-and-flag approach over a more ambitious hybrid reordering fix because it is the lower-risk option, and because it is consistent with this paper’s own three-tier philosophy of flagging uncertainty honestly instead of guessing at an answer we cannot actually verify.

A related, tested-but-inconclusive hypothesis. We also tested a related hypothesis directly instead of leaving it as an untested guess: that environment-dependent branching, meaning checks such as `torch.cuda.is_available()` whose outcome depends on the machine the code happens to run on rather than anything visible in the file itself, might be a separate static-analysis blind spot distinct from the `execution_count`-sort issue described above. The result of this test was genuinely inconclusive, and we report it as such. 8 of the 41 items in our earlier ground-truth snapshot matched an environment-branching pattern, and the raw correlation looked striking at first glance: an 88.2% wrong-answer rate for these flagged items, compared with 48.1% for items without the flag. However, every one of those 8 flagged items shares the identical `phantom_in_visual/ec_order_value=None` signature already fully explained by the `execution_count`-sort mechanism above, so this particular corpus contains zero cases that could actually separate the two hypotheses from each other. We report this honestly as an inconclusive result: it is not evidence that the environment-branching hypothesis is false, only evidence that this specific corpus happens to be unable to test it one way or the other. We think a tested-but-inconclusive hypothesis is still worth keeping in the record instead of quietly dropping it. (`reports/environment_branching_check.md`)

6.2 Internal validity

At $n=52$, after fixing the judge bug identified above and replicating the result across $13 \rightarrow 41 \rightarrow 52$ items, `ours` shows no accuracy-when-answered advantage over `raw` or over the plain compression control (`plain_strip`): `ours` and `plain_strip` are tied at 27%, and `raw` trails only nominally at 23%, all within noise. Because the same 52 items are scored under all three conditions, we also report a paired risk difference rather than only comparing separate confidence intervals: `ours` beats `raw` on 6 items and loses to it on 4, for a paired risk difference of +3.8 percentage points (95% CI $[-8.0, +15.7]$); `ours` beats `plain_strip` on 3 items and loses to it on 3, for a paired risk difference of exactly 0.0 percentage points (95% CI $[-9.2, +9.2]$). Both intervals contain zero by a wide margin, which is a more direct way of saying the same thing as the overlapping Wilson intervals above: at this sample size, we cannot distinguish `ours` from either baseline on a per-item basis. The real-notebook claim this paper actually stands behind rests entirely on reach and coverage, not on any demonstrated per-item accuracy advantage of structure over compression. This is a further downgrade from an already-downgraded 13-item and then 41-item pre-fix draft, and

we report it transparently as such: this result got smaller and more modest at every stage we checked it, not larger. We did not carry out a formal power analysis before concluding that this is a genuine null result rather than simply an underpowered one; at the effect sizes we actually observed, a substantially larger item count, plausibly 150 or more, would be needed to resolve the question with real confidence, and we flag this explicitly as an open question instead of treating our current $n=52$ as the final word. Real-notebook state accuracy in this paper is therefore a 52-item case study covering the full population we could recover across a Python 3.14 environment with torch and torchvision, plus a separate Python 3.12 environment with TensorFlow, not a random sample of notebooks in general. One TensorFlow notebook remains unexecutable because of a flaky, timeout-sensitive network download rather than a missing dependency, and we observed run-to-run item-count variance of 52 to 55 due to genuinely network-dependent code inside the notebooks themselves. Finally, static extraction cannot detect the runtime failure of code that looks syntactically valid; these phantom-variable cases are, in fact, the dominant remaining source of error regardless of which serialization condition is used.

6.3 External validity

Fifteen answer models in total were tested on the synthetic stateful benchmark: nine are from OpenAI, and six (NVIDIA Nemotron-3-Super-120B, MiniMax-M3, Cohere North-Mini-Code, InclusionAI Ling-3.0-Flash-Fin, Poolside Laguna-S-2.1, and Zhipu GLM-5.3-Flash, each accessed through its own provider’s free tier) are not. The non-OpenAI models ran a reduced grid of one length by three depths by one or two trials by three conditions, sized to fit inside each provider’s rate limit. **ours** scores 100% on all six of these models regardless; five of the six reproduce the qualitative pattern of imperfect, non-monotonic **raw** accuracy seen throughout the OpenAI curve, while the sixth (GLM-5.3-Flash) scored 100% raw on only three trials. We consider this reassuring, but three to six trials per non-OpenAI model on a reduced grid is not enough to claim broad cross-vendor generality on its own; it tells us that most of these six additional vendors did not contradict the pattern, not that we have demonstrated general cross-vendor independence, and the one exception is too thinly sampled to interpret in either direction. Two further free-tier candidates, Google Gemma-4-31B and NVIDIA Nemotron-3-Ultra-550B, along with a Gemini 3.7/3.8 Flash attempt, were tried and then dropped after we confirmed persistent upstream unavailability: exhausted retries in the first two cases, and repeated HTTP 400 and 503 errors traced specifically to free-tier capacity limits (not to our own account, since we confirmed this was reproducible with two independent API keys) in the Gemini case. This follows our project’s standing policy of dropping a source that has proven consistently unreliable rather than continuing to fight it or reporting a data point we cannot trust. The real-notebook evaluation used gpt-4o exclusively. All of our real-notebook data comes from a single benchmark corpus, JunoBench, so we have not tested how well our findings generalize to notebooks outside this specific crashed-machine-learning population. On the cost of Tier 3, our runtime-verification tier (Sect. 2): we already restrict this tier’s use to settings where a live kernel is genuinely available, and the outside evidence we found supports keeping that restriction and does not suggest it is overly cautious. A 2026 preprint (FlowBook) reports lightweight dynamic instrumentation with a median overhead of 70 milliseconds per cell, which is cheap, but only while a notebook is *already executing*. That is not a way to validate one isolated cell in the absence of a kernel, so it does not relax our own no-kernel constraint for Stage 1; it only suggests that Tier 3 is less costly than we might otherwise assume in settings, such as an interactive agent session, where a kernel is already attached for some other reason.

System	Task	Live kernel?	Serialization a variable?	Headline result
CRABS (Li et al., 2025a)	Execution-dependency graph reconstruction	No; LLM disambiguation step instead	No; fixed input format	98–99% F1 on 50 curated Kaggle notebooks
LongDS-Bench (Xu et al., 2026)	Long-horizon analytical state maintenance	Yes; agent executes tools	No	Best model 48.45% accuracy; ~47-pt drop late in session
DSAgentBench (Rahman et al., 2026)	End-to-end data-science workflow completion	Yes; real computer environment	No	Best model 56.70% task success
marimo (marimo, 2026)	Reactive execution, no serialization step	Yes, by design	N/A; format itself is the fix	40/40 tooling-reliable, 40/40 blocked by missing deps/data (this paper’s test)
This paper	State-tracking under a controlled serialization change	No; static, zero-LLM-cost	Yes; the independent variable	100% synthetic, null real-notebook accuracy, coverage gain confirmed

Table 4: How the closest prior systems compare to this paper along the dimensions that matter for our claim. None of the four vary serialization format as a controlled experimental variable the way we do; three of the four require a live kernel or agent-executed environment that our zero-LLM-cost, no-kernel setting is specifically designed to avoid needing.

7 Related Work

Several notebook agents and stateful runtimes, including DatawiseAgent (You et al., 2025), CaveAgent (Ran et al., 2026), DeltaBox (Dong et al., 2026), and I-PTC (Gonzalez, 2026), all assume a live kernel is available. Serialization-side tooling that does not assume a kernel tends to be lossy: Jupyterxtext (Jupyterxtext, 2026) and notebookllm (notebookllm, 2026) both strip outputs entirely, and existing base64-stripping tools have not, to our knowledge, been formally evaluated for their effect on model accuracy. Existing benchmarks, including Themisto (Grotov and Titov, 2025), JunoBench (Wang et al., 2025), NbQA/Jupiter (Li et al., 2025b), and DABStep (Egg et al., 2025), test downstream tasks rather than serialization formats themselves; none of them isolate execution-order reconstruction as an independent variable the way we do here. As far as we are aware, no prior work measures LLM state accuracy against dual-order executable ground truth on real notebooks. Table 4 summarizes, for the four prior systems discussed in detail below, what each one actually measures and how that differs from our own evaluation.

The closest prior work to ours is **CRABS** (Li et al., 2025a) (COLM 2025), which reconstructs a notebook’s execution-dependency graph using static and syntactic analysis combined with an LLM disambiguation step, reporting 98–99% F1 on 50 real Kaggle notebooks. This is meaningful evidence that a static-analysis-plus-LLM approach can recover a notebook’s execution structure without actually re-executing it. CRABS’s accuracy, however, depends on that LLM step, which places it closer to our optional Stage 2 than to our zero-LLM-cost Stage 1 taken on its own; its evaluation sample of “highly-upvoted” Kaggle notebooks is also noticeably cleaner than the crashed, messy corpus we study here, so we do not know whether CRABS’s accuracy would transfer to a harder

setting like ours. Two more recent benchmarks independently support the broader motivation for this work, beyond the older reproducibility literature we already cite: **LongDS-Bench** (Xu et al., 2026) finds that even the best model reaches only 48.45% accuracy at maintaining and restoring analytical state across long-horizon tasks derived from real Kaggle notebooks, with a roughly 47-point accuracy drop from early to late turns that the authors attribute to a genuine failure to maintain state rather than simply running out of interaction budget; and **DSAgentBench** (Rahman et al., 2026) finds that the best model succeeds on only 56.70% of 275 real, full-environment data-science tasks. Neither benchmark targets serialization directly, but both reinforce our motivating claim that state-tracking during notebook-style analysis remains an unsolved problem at the level of whole agent systems, not simply a niche concern within the older reproducibility literature.

A structurally different alternative to our approach is **reactive, or dataflow, notebooks such as marimo** (marimo, 2026), which eliminate hidden state and out-of-order execution by design instead of by reconstructing it after the fact, at the cost of requiring a conversion away from the standard `.ipynb` format. We tested marimo’s own official conversion tooling (`marimo convert` followed by `marimo export html`) on the same 40-notebook real, crashed corpus used in Sect. 5.1. Conversion and reactive execution never crashed or timed out on any of the forty notebooks, but every single one of them had at least one cell fail during execution, almost always because of missing local data files or third-party packages not present in the runtime environment, which is the same “missing dependency or data” failure mode our own dual-order re-execution ground truth already reports for these same notebooks. We read this as evidence *in favor of*, not against, a static, execution-free approach on the specific question of reach: any method that requires live execution, marimo’s reactive model included, inherits the same real-world unavailability of a notebook’s original data and environment, while a purely static method cannot be blocked by a missing file or package because it never attempts to run anything at all. We did not compare marimo’s own recovered variable states against our executable ground truth in the cases where both approaches actually can execute successfully; that comparison remains open for future work. We emphasize that this experiment tested only reach and executability, not comparative state accuracy, and it should be read with that scope in mind rather than as a claim that our static approach is more accurate than marimo’s reactive one. (`reports/marimo_conversion_report.md`)

8 Conclusion

This paper is built around two defects and one underlying mechanism: strip what the file wastes, reconstruct what it scrambles, and be explicit about what cannot actually be known. In a fair, controlled synthetic setting, execution-order reconstruction is both necessary and sufficient to fix the measurable state-tracking failure we identified, at roughly 95% fewer tokens and effectively no added inference-time cost. That result does not, however, transfer cleanly to real notebooks as an accuracy advantage. On 52 real order-divergent variables from 16 crashed machine-learning notebooks, using gpt-4o as the sole answer model, our serializer and a plain compression control are statistically tied, and the raw file trails both, all within noise once an answer is actually reached, and this conclusion held when we expanded the corpus from 41 to 52 items rather than reversing itself. We want to be precise about what this null result does and does not license: it is evidence against an accuracy advantage for gpt-4o on this specific, deliberately hard population of crashed ML notebooks, not a general claim about every model or every kind of notebook. Whether the same null result holds for other answer models, or for calmer notebook populations such as teaching or exploratory data analysis, remains untested and is the most direct follow-up this paper’s own evidence calls for. What does transfer, and what we are prepared to stand behind, is narrower but

still substantial: raw JSON is frequently unusable on real notebooks at all, overflowing context on 62% of real divergent items, and this gap widened instead of narrowing as we added more real notebooks to the corpus, while a deterministic, zero-LLM-cost serializer keeps the notebook readable almost every time. On real, messy data, then, a serializer that is honest about what it does not know functions first as a coverage tool, and only potentially later as an accuracy tool. Even under close scrutiny of our own pipeline, including a judge bug we found and fixed and an edge case in our own reordering mechanism that we also found and fixed, this particular coverage claim held up.

For someone building a notebook agent today, we would summarize the practical guidance as follows. Deploy JupPreprocessor when notebooks are bloated, out of order, or otherwise at risk of overflowing context, since that is where the coverage gain is real and, on the evidence gathered here, essentially free. Do not expect an accuracy improvement on notebooks that are already lean and already in execution order; on such notebooks, added structure can hurt rather than help, as the Themisto result shows. Treat the `execution_order_risk` flag as a signal to fall back to a more conservative, uncertainty-preserving reading of the affected cells, not as confirmation that anything is actually broken; it fires on a real but specific pattern (a cell re-run late in a session) that neither our tool nor a live re-execution can fully resolve without more information than the file itself contains.

Statements and Declarations

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Author contributions. Single-author manuscript: MD. Raiyan Bin Rafique conceived the study, designed and ran all experiments (with the LLM-assistance disclosed in Sect. 3.1), analyzed the results, and wrote the manuscript.

Ethical approval. This study did not involve human participants, animal subjects, or personally identifying data; no ethical approval was required. All notebook data is drawn from JunoBench, a publicly released, pre-existing benchmark of crashed machine-learning notebooks (Wang et al., 2025).

Data availability. The complete source code for JupPreprocessor, the evaluation harnesses (Notebook-NIAH, the real-notebook dual-order ground-truth extractor, the Themisto and JunoBench evaluation scripts), all measured reports underlying every quantitative claim in this manuscript, and the full project changelog documenting the correction history described in Sect. 5.4–Sect. 5.5 are available at <https://github.com/hodini007/notebook-research>. EMSE’s editorial policy states that submissions missing required declarations will be returned as incomplete.

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